

CH7 數據分析

$$1.66 \quad 2.(1)71 \quad (2)67 \quad 3.-20\% \quad 4.8 \quad 5.\frac{1}{30}$$

$$6.-0.85 \quad 7.0.875 \quad 8.r_2 < r_5 < r_3 < r_4 < r_1 \quad 9.0.3$$

$$10.(1) \quad 0.8 \quad (2)y = \frac{2}{5}x - \frac{86}{5} \quad (3)52$$

解析

$$\begin{aligned} 1. \bar{X} &= \frac{70 \times 30 + 60 \times 20}{30 + 20} = \frac{70 \times 3 + 60 \times 2}{3 + 2} \\ &= 60 + \frac{10 \times 3 + 0 \times 2}{3 + 2} = 66 \end{aligned}$$

$$2.(1) \text{共有 } 1 + 2 + 3 + \cdots + 100 = \frac{100 \cdot 101}{2} = 5050 \text{ 個}$$

其中位數為第 2525 及第 2526 個之平均值

$$\text{又 } 1 + 2 + \cdots + n = \frac{n(n+1)}{2} \Rightarrow 2525$$

$$\text{取 } n = 70, \frac{70 \cdot 71}{2} = 2485; n = 71, \frac{71 \cdot 72}{2} = 2556$$

$\therefore$ 第 2525 及 2526 項均為 71 $\therefore$ 中位數 = 71

$$(2) \bar{X} = \frac{1 \cdot 1 + 2 \cdot 2 + \cdots + 100 \cdot 100}{1 + 2 + 3 + \cdots + 100} = \frac{1^2 + 2^2 + \cdots + 100^2}{1 + 2 + \cdots + 100} = \frac{\frac{1}{6} \cdot 100 \cdot 101 \cdot 201}{\frac{1}{2} \cdot 100 \cdot 101} = \frac{201}{3} = \underline{67}$$

$$\begin{aligned} 3. & \sqrt[3]{(1 + 60\%)(1 - 20\%)(1 - 60\%)} - 1 \\ &= \sqrt[3]{0.512} - 1 = 0.8 - 1 = -0.2 \end{aligned}$$

故為-20%

$$4. \text{原標準差 } \sqrt{\frac{x_1^2 + x_2^2 + \cdots + x_{39}^2 + 40^2}{40}} - 51^2 = \sqrt{3}$$

$$\Rightarrow x_1^2 + x_2^2 + \cdots + x_{39}^2 = 102560$$

$$\text{新平均數 } \mu = \frac{51 \times 40 - 40}{40} = 50$$

$$\Leftrightarrow \text{新標準差 } \sigma = \sqrt{\frac{x_1^2 + x_2^2 + \cdots + x_9^2 + 0^2}{40}} - 50^2 = \underline{8} \text{ (分)}$$

$$5. z_1 - z_2 = \frac{85-72}{15} - \frac{90-86.5}{4.2} = \frac{13}{15} - \frac{3.5}{4.2} = \frac{1}{30}$$

$$6. r(U, V) = r(4x + 3, -7y + 1) = -r(x, y) = \underline{\underline{-0.85}}$$

$$7. \mu_x = \frac{135}{5} = 27, \mu_y = \frac{105}{5} = 21$$

$$r = \frac{2842 - 5 \times 27 \times 21}{\sqrt{3661 - 5 \times 27^2} \times \sqrt{2209 - 5 \times 21^2}} = \frac{7}{\sqrt{16} \times \sqrt{4}} = 0.875$$

8. 由題圖知， r_1 接近 1， r_2 接近 -1， r_3 接近 0，

而 $0 < r_4 < 1, -1 < r_5 < 0$

$$\therefore r_2 < r_5 < r_3 < r_4 < r_1$$

$$9. \text{迴歸直線 } y - \mu_y = r \cdot \frac{\sigma_y}{\sigma_x}(x - \mu_x)$$

$$\Rightarrow y - 5 = 0.6 \cdot \frac{\sigma_y}{\sigma_x}(x - 3) \text{ 過點 } (1, 1)$$

$$\Rightarrow 1 - 5 = 0.6 \cdot \frac{\sigma_y}{\sigma_x}(1 - 3) \therefore \frac{\sigma_x}{\sigma_y} = 0.3 \quad 10. \mu_X = \frac{840}{5} = 168, \mu_Y = \frac{250}{5} = 50$$

	A	B	C	D	E	總和
x_i	160	164	168	172	176	
y_i	48	46	50	54	52	
$x_i - \mu_x$	-8	-4	0	4	8	
$y_i - \mu_y$	-2	-4	0	4	2	
$(x_i - \mu_x)^2$	64	16	0	16	64	$S_{xx} = 160$
$(y_i - \mu_y)^2$	4	16	0	16	4	$S_{yy} = 40$
$(x_i - \mu_x) \times (y_i - \mu_y)$	16	16	0	16	16	$S_{xy} = 64$

$$(1) r_{(X,Y)} = \frac{S_{XY}}{\sqrt{S_{XX}\sqrt{S_{YY}}}} = \frac{64}{\sqrt{160}\sqrt{40}} = 0.8$$

$$(2) \ y - \mu_Y = \frac{S_{XY}}{S_{XX}}(x - \mu_X) \Rightarrow y - 50 = \frac{64}{160}(x - 168)$$

$$\Rightarrow y = \frac{2}{5}x - \frac{86}{5}$$

$$(3) \text{當 } x = 173 \text{ 時，推測 } y = \frac{2}{5} \times 173 - \frac{86}{5} = \underline{52} \text{ (公斤)}$$